A Catalan clique partition of middle Johnson graphs at powers of two

Alex Chan
Harvard University and National Bureau of Economic Research
achan@hbs.edu

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Abstract

For every power of two $k \geq 2$, we partition the vertices of the middle Johnson graph $J(2k, k)$ into cliques of size $k + 1$. Consequently its vertex clique-covering number, or equivalently the chromatic number of the complementary generalized Kneser graph, is the kth Catalan number. A single binary-syndrome rule gives the partition and a matching clique bound proves optimality. This establishes one infinite family within Jensen–Toft Problem 9.13; it does not determine the chromatic numbers for general parameters. The result supplies the numerical consequence of a recent conjecture concerning constant-weight lexicodes, without making a claim about the lexicodes themselves.

Keywords. Generalized Kneser graph; Johnson graph; clique partition; Catalan number; binary syndrome.

1 Statement and context

For $0 < t < k < n$, the generalized Kneser graph $K(n, k, t)$ has the k-subsets of an n-element set as its vertices; distinct sets are adjacent when their intersection has size less than t. Jensen and Toft ask for its chromatic number in Problem 9.13 of [1]. The Johnson graph $J(n, k)$ has the same vertices, with adjacency when the intersection has size $k - 1$. In particular,

$$K(n, k, k-1) = \overline{J(n, k)}, \qquad \chi(K(n, k, k-1)) = \theta(J(n, k)), \tag{1}$$

where θ is the minimum number of vertex cliques needed to cover a graph.

Write $C_k = \frac{1}{k+1}\binom{2k}{k}$ for the kth Catalan number. Jørgensen [2] established $\theta(J(2k, k)) = C_k$ for $k = 1, 2, 4, 6, 8, 16$ and proposed a lexicode property for powers of two whose conclusion includes this equality. The construction below proves the equality for the entire power-of-two family. It does not assert that the lexicode has the proposed form, or address other values of k. The first case beyond those values is $k = 32$, for which the formula gives $\chi(K(64, 32, 31)) = C_{32} = 55{,}534{,}064{,}877{,}048{,}198$.

Theorem 1. *For every integer $r \geq 1$ and $k = 2^r$,*

$$\chi\big(K(2k, k, k-1)\big) = \theta\big(J(2k, k)\big) = \frac{1}{k+1}\binom{2k}{k} = C_k.$$

An optimal coloring and an optimal clique partition are given explicitly by (2) *below.*

2 Syndrome partition and optimality

Put $m = r + 1$ and label the $2k = 2^m$ ground elements by the vectors of $X = \mathbb{F}_2^m$. For a subset $A \subseteq X$, define its syndrome by

$$\sigma(A) = \bigoplus_{a \in A} a \in X.$$

Since the ground elements are labeled by *all* vectors of X, the vector $\sigma(A)$ also names a unique ground element. For every k-subset S set

$$c(S) = S \triangle \{\sigma(S)\}. \tag{2}$$

This is the color assigned to S; as a Johnson-graph construction, it is the center of the clique containing S. It has size $k - 1$ or $k + 1$, and $\sigma(c(S)) = 0$.

For any zero-syndrome set T of either of those sizes, let

$$\mathcal{F}_T = \{S \subseteq X : |S| = k,\ |S \triangle T| = 1\}.$$

If $S \in \mathcal{F}_T$ and $S \triangle T = \{x\}$, then $\sigma(S) = \sigma(T) \oplus x = x$. Thus $c(S) = T$. Conversely, every k-set S belongs to $\mathcal{F}_{c(S)}$. The nonempty $\mathcal{F}_T$ are therefore *exactly* the fibers of c and partition all $\binom{2k}{k}$ vertices.

If $|T| = k - 1$, the fiber consists of all k-sets containing T, and has $|X \setminus T| = k + 1$ members. If $|T| = k + 1$, it consists of all k-sets contained in T, and again has $k + 1$ members. In either case, every pair of distinct members intersects in $k - 1$ elements. Thus every fiber is a clique of $J(2k, k)$ and an independent color class of $K(2k, k, k - 1)$. This gives

$$\theta(J(2k, k)) = \chi(K(2k, k, k - 1)) \leq \frac{1}{k + 1}\binom{2k}{k}. \tag{3}$$

For completeness, no clique of $J(2k, k)$ has more than $k + 1$ vertices. Indeed, choose distinct members $A = R \cup \{a\}$ and $B = R \cup \{b\}$ of a clique, where $|R| = k-1$. Every other member containing R lies in the star $\{R \cup \{x\} : x \in X \setminus R\}$. Every member D not containing R must intersect both A and B in $k - 1$ elements, and hence has the form $D = (R \setminus \{z\}) \cup \{a, b\}$ for some $z \in R$. If such a D exists, a member $R \cup \{x\}$ of the clique must have $x \in \{a, b\}$, since otherwise its intersection with D has size $k - 2$. The entire clique then lies in the top $\{(R \cup \{a, b\}) \setminus \{x\} : x \in R \cup \{a, b\}\}$. A clique of at most one member satisfies the bound as well. Both the star and the top have $k + 1$ members. Thus any clique cover needs at least $\binom{2k}{k}/(k + 1)$ cliques; by (1) the same is a lower bound on the number of colors. Together with (3), this proves Theorem 1. □

3 The centers as a constant-weight code

The zero-syndrome $(k - 1)$-subsets form a code

$$\mathcal{C} = \{T \subseteq X : |T| = k - 1,\ \sigma(T) = 0\}.$$

It has $C_k/2$ members. Indeed $k - 1$ is odd, so translation by $v \in X$ takes the syndrome of T to $\sigma(T) \oplus v$. All $2k$ syndrome classes have equal size, and hence

$$|\mathcal{C}| = \frac{1}{2k}\binom{2k}{k - 1} = \frac{1}{2}C_k.$$

Two distinct members of $\mathcal{C}$ cannot differ by one exchange: the two exchanged elements would have equal labels. They also cannot be disjoint. For the latter assertion, $\bigoplus_{x\in X} x = 0$ since $m \geq 2$; if disjoint $T, U \in \mathcal{C}$ left exactly two elements a, b outside $T \cup U$, then $a \oplus b = 0$, contrary to $a \neq b$. Complements of the members of $\mathcal{C}$ are precisely the zero-syndrome $(k+1)$-subsets. Consequently the partition above has $C_k/2$ stars and $C_k/2$ tops, paired by complementation.

More explicitly, the cover contains the star with center T if and only if it contains the top with center $X \setminus T$. Thus it has the complement symmetry considered in [2, Remark 4.17] for each power-of-two case in Theorem 1.

The minimum-distance and intersection properties of $\mathcal{C}$ explain its relation to the code-based method of [2]. The proof here uses the explicit syndrome class, and does not identify it with any lexicographically greedy code. It proves the stated special family of Jensen–Toft 9.13, while the problem for unrestricted (n, k, t) remains outside its scope.

Data and code availability

The chromatic-number formula in Theorem 1 has also been formally verified in Lean 4 with mathlib.[1] The Lean source, pinned dependencies, Python graph generator, and finite witness for $K(8, 4, 3)$ are available at `https://github.com/alexgocardinal/jt913-xor-coloring`. The mathematical proof above does not rely on the finite computations.

References

[1] T. R. Jensen and B. Toft, *Graph Coloring Problems*, John Wiley & Sons, New York, 1995, Problem 9.13, pp. 161–162. doi:10.1002/9781118032497.

[2] S. F. Jørgensen, On the clique covering numbers of Johnson graphs, *Designs, Codes and Cryptography* **93** (2025), 3689–3705. doi:10.1007/s10623-025-01663-3.

[1] The formalization includes the XOR coloring, fiber sizes, independent-set bound, exact chromatic number, and transfer to the conventional ground labels. The Johnson-graph reformulation and the supplementary constant-weight-code properties in Section 3 are not separately formalized.